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<u>Title / Título</u> PROGRAM FREM USER'S MANUAL				
<u>Keywords / Palabras clave</u> SOFTWARE, INTER-RIVET BUCKLING			Access class Clasificación acceso Airbus Amber P1 Revisions Record Registro de Revisiones Pag. 2	
<u>Summary / Resumen</u> The purpose of this Technical Report is to define the way to use the FREM program. This program calculates the allowable stress for inter-rivet buckling in metallic skins. This revision of the user's manual corresponds to FREM version 3.2. In this version of the program the calculation of the effective skin width has been removed, because, for a consistent calculation, the stress in the stringer is required, and actually it is not part of the input data of the program. See appendix for further clarifications in case the user is interested in this topic.				
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1 INTRODUCTION

1.1 TECHNICAL SHEET OF THE PROGRAM

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PROGRAM
DATE

FREM

06/2020

OBJECTIVE

This program calculates the allowable stress for inter-rivet buckling in a metallic panel under compression loads.

LANGUAGE
DEVELOPMENT TOOLS
OS/PLATFORM

Fortran 77

MS Visual Studio 2013

Linux (RHEL) and Windows (x86)

Visual Basic Net

MS Visual Studio 2013

Windows (x86)

NUMBER OF LINES (APPROX.)
EXECUTION MODE

2200 (Fortran, analysis core)

Console (input data in text file)

+ 1700 (Visual Basic .Net)

Interactive GUI

Total 3900 (approx..)

Notes:

- Blank lines and commented lines excluded
- Count for Visual Basic code excludes lines created by Visual Studio in *designer.vb* files

AUTHOR
VERSION

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1.0

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2.0 - 3.1

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3.2

ADDITIONAL DOCUMENTS

See list of applicable references in chapter §1.3.

1.2 SOFTWARE PURPOSE AND DESCRIPTION

This report makes up the User's Manual for version 3.2 of the FREM Program, which has been developed by Airbus DS / EADS-CASA from the year 2001 to date.

This program calculates the allowable stress for inter-rivet buckling in metallic skins.

This revision of the user's manual corresponds to FREM version 3.2. In this version of the program the calculation of the effective skin width has been removed, because, for a consistent calculation, the stress in the stringer is required, and actually it is not part of the input data of the program. See appendix for further clarifications in case the user is interested in this topic.

1.3 ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
GUI		Graphical User Interface
I/O		Input and Output
MSVS		Microsoft Visual Studio 2013 (plus Intel Visual Fortran Compiler version 14 and IMSL Fortran Numerical Library version 6.0.0)

1.4 REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	NT-T-ADP-08013	Is. B	Theoretical Manual of the FREM Program.
Ref. 2	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 3	[M. C. Y. Niu]	2 nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 4	[NACA] TN 3781	1957	HANDBOOK OF STRUCTURAL STABILITY. PART I – BUCKLING OF FLAT PLATES.

1.5 DIFFERENCES AMONG VERSIONS

- Version 1.0:

Initial version developed in Fortran 77/90 for the Sun Solaris computer with a graphic user interface developed in Java.

- Version 2.0:

The user can choose the language: Spanish or English.

The program is restructured to allow the creation of a dynamic link library DLL for Windows operative system.

The previous graphic user interface in Java is discarded and substituted by other developed in Visual Basic .NET for Windows.

Validation examples are created for further versions of the program.

- Version 3.0:

The Ramberg-Osgood number was one of the program inputs. Now, it is calculated from the maximum allowable tensile stress, F_{tu} , the yield stress under compression, F_{cy} , and the plastic deformation for F_{tu} , e .

Furthermore, the Ramberg-Osgood number, the tangent and secant modulus corresponding to the allowed stress for inter-rivet buckling and the minimum effective width of the panel is included between the output results.

- Version 3.1:

General update of the of the program.

- Version 3.2:

- ✓ Solaris/Unix version deleted
- ✓ Functionality to return the effective width deleted
- ✓ Software documentation updated accordingly

2 PROGRAM EXECUTION

2.1 WINDOWS/PC VERSIONS

The Visual Basic executable program Frem.exe (as well as an electronic copy of this manual and the program examples) is located within this folder in Airbus DS - Getafe - IT network:

```
\\multivac\InfoCalculo\Soft\Frem
```

To execute the program, the user has to click twice on the Visual Basic executable program.

There is another executable file with the same name, which is the Fortran program in console mode, in the subdirectory .\Consola within that folder (and another copy of the console-mode executable in \\multivac\InfoCalculo\Soft\Consola, this last path than can be added to the Windows system PATH to facilitate its use).

Execution in the command line terminal

This program can be executed by typing its full path name. If the software symbol table has been previously assigned (consult with the System Manager in order to know how to assign this symbol table), it can be executed by typing the command:

```
frem
```

The program asks for the language to use and all required data to perform the analysis. The results will be presented on the screen. Additionally it is possible to specify an output file, as shown in the following example:

```
*****
*                                     *
*          <<< F R E M >>>          *
*                                     *
*****
```

Idioma / Language:

=====

1 ... ESPAÑOL

2 ... ENGLISH

Idioma / Language: 2

```
.....
.....
.....
```

Do you want to print the results in an file? (Y/N): Y

Name of the output results file: Example.txt

Details of the interactive data flow on the screen are provided in the examples included in this report. The CPU time for calculation is negligible.

2.2 LINUX/HPC VERSION

To run interactively this version of the program (console application), you must have already done the assignment of the symbols list in Airbus DS Structural Analysis Office using the command:

```
. asignaciones_calculo
```

This instruction can be executed in the commands terminal prior to execute the program, but it is strongly recommended to include this statement in the files ".profile", and ".kshrc" (if working in Korn Shell) or ".bashrc" (if you work in Bash Shell). To do this, the following lines should be included in the file ".profile":

```
. asignaciones_calculo
ENV=$HOME/.kshrc
export ENV
BASH_ENV=$HOME/.bashrc
export BASH_ENV
```

In files ".kshrc" and ".bashrc" the following line shall be added:

```
. asignaciones_calculo
```

The instruction to run the program in interactive mode in the HPC/CAE Linus environment is:

```
frem
```

... with a sequence of questions to user similar to the Windows PC version shown in previous page.

3 INPUT DATA

3.1 WINDOWS PC/GUI

An application has been developed in Visual Basic for Windows environment. The original routines of the program for calculation and output results have been included in a Dynamic Link Library, which is used by this application. Therefore the same results will be obtained, with only differences due to rounding.

Since this is a Windows application, it must be installed in the user's PC before it can be executed. For installation, please consult with the system manager.

The application can be executed by "double click" in the file Frem.exe, located in the PC's server, or by means of a link in the "desktop" or in the "start menu".

The input data must be typed in the application form. When all data have been introduced, the results are automatically presented.

The form, after loading the data corresponding to the example 2 (see section §5), has the following appearance:

Frem

Material:

Denomination 7050 T7451

Type of material Clad

Mechanical properties of the core:

Young modulus in compression E_c 73084.0

Ultimate tensile stress F_{tu} 524.0

Yield stress in compression F_{cy} 441.3

Poisson's modulus μ 0.33

Plastic deformation for ultimate stress (%) 10.0

Number of Ramberg-Osgood in compression 22.78

Mechanical properties of the cladding:

Yield stress in compression F_{cyc} 75.0

Percentage of thickness of cladding 20.00

Fastening data:

Panel thickness at the fastened zone 1.5

Fasteners pitch 30

Type of fasteners head Protruding

Clamping coefficient of fasteners 3.0

Force units: N

Length units: mm

Stress units: N/mm²

Language: ☐ Español ☒ English

Buckling stress in plastic regime -310.8

Tangent modulus for the buckling stress 50387.1

The problem has been solved without errors

Figure 1. Input data form.

The buttons located at the right side of the form are to write the results in a file (upper button) or to write the results in a printer (lower button).

3.2 INPUT FILE

All input data for the program have to be provided in interactive mode. They are enumerated and explained below:

- **Language:**

The user can select the language, Spanish or English. All interactive dialogs as well as the output results file will be written in the selected language. The question is made in the following screen:

```
*****
*                                     *
*          <<< F R E M >>>          *
*                                     *
*****
```

```
Idioma / Language:
=====
```

- 1 ... ESPAÑOL
- 2 ... ENGLISH

```
Idioma / Language: 2
```

- **Material data:**

The program requests the material denomination (only for information in output file) and if it is clad or unclad, as shown below:

```
*****
*                                     *
*   MATERIAL   *
*                                     *
*****
```

```
Material denomination ..... 7050 T7451
```

```
TYPE OF MATERIAL:
=====
```

- 1 ... Homogeneous (unclad)
- 2 ... Clad

```
Type of material ..... 2
```

- **Units:**

The user can select the units for forces and lengths in the following way:

```
*****
*               *
*   UNITS      *
*               *
*****
```

FORCE UNITS:

=====

- 1 ... N
- 2 ... daN
- 3 ... kN
- 4 ... kg
- 5 ... lb
- 6 ... klb

Force units 1

LENGTH UNITS:

=====

- 1 ... mm
- 2 ... cm
- 3 ... m
- 4 ... in

Length units 1

The above units will be used only for information in the output results. Units for elasticity modulus and stresses will be a combination of the above selected units.

- **Mechanical properties of the material:**

If the material is unclad, the program asks for its mechanical properties. In case of clad material, the program asks for mechanical properties of the core and the cladding, as follows:

- ✓ Homogeneous material (unclad):

```
*****
*
*   MECHANICAL PROPERTIES OF THE MATERIAL   *
*
*****
```

Young modulus in compression 73084

Ultimate tensile stress 524.0

Yield stress in compression 441.3

Poisson's modulus 0.33

Plastic deformation for ultimate stress (%) 10.00

Number of Ramberg-Osgood in compression 22.78

✓ Clad material:

```
*****
*
*   MECHANICAL PROPERTIES OF THE CORE   *
*
*****
```

Young modulus in compression 73084

Ultimate tensile stress 524.0

Yield stress in compression 441.3

Poisson's modulus 0.33

Plastic deformation for ultimate stress (%) 10

Number of Ramberg-Osgood in compression 22.78

```
*****
*
*   MECHANICAL PROPERTIES OF THE CLADDING   *
*
*****
```

Yield stress in compression 75.0

Percentage of thickness of cladding 20

In both cases, the number of Ramberg-Osgood is automatically displayed as calculated from its corresponding expression (see the theoretical manual for this purpose).

- **Fastening data:**

The details of the fastening are input as follows:

```
*****
*                               *
*   FASTENING DATA           *
*                               *
*****
```

Panel thickness at the fastened zone 1.5

Fasteners pitch 30.0

TYPE OF FASTENERS HEAD:

=====

- 1 ... Countersunk head
- 2 ... Protruding head
- 3 ... Traction head
- 4 ... Other kind of head

Type of fasteners head 2

Clamping coefficient of fasteners 3.0

When the user selects a specific type of fasteners head, countersunk, protruding or traction head, the program does not ask for the clamping coefficient and displays it automatically, assuming the following values:

Head	Clamping coefficient
Countersunk	1.5
Protruding	3.0
Traction	4.0

4 OUTPUT RESULTS

The FREM program provides the analysis results on the screen, in interactive way. After the input data, the following results are displayed...

Final results:

Displayed as follows:

```
*****
*
*   ALLOWABLE STRESS FOR INTER-RIVET BUCKLING   *
*
*****
```

Buckling stress in plastic regime -334.1 N/mm²

Tangent modulus 50387.1 N/mm²

4.1 OUTPUT FILE

It is possible to keep the whole set of input data together with the above output results in a text file. For this purpose, the program asks if the user wants to save the results in a file.

Do you want to print the results in an file? (Y/N): Y

Name of the output results file: Example.txt

4.2 WARNING AND ERROR MESSAGES

When the program detects an error in the input data, it writes the corresponding message in the console and repeats the whole input data process. The error messages that might appear in the console, are listed below:

Error messages. English output:

ERROR: Selection out of bounds.

ERROR: No numerical value.

ERROR: The Young modulus must be positive.

ERROR: The ultimate tensile stress must be greater than zero.

ERROR: The yield stress must not be zero.

ERROR: The yield stress must be lower than the ultimate stress.

ERROR: El módulo de Poisson tiene que ser positivo.

ERROR: La deformación plástica para esfuerzo último tiene que ser mayor que $(F_{tu}/F_{cy}) \cdot 0.2 \%$.

ERROR: The number of Ramberg-Osgood must be: $N_{ro} > 1$.

ERROR: The percentage of thickness of cladding must be: $0 < R_p < 100$.

ERROR: The thickness must be positive.

ERROR: The pitch must be positive.

ERROR: The clamping coefficient must be positive.

ERROR: The interval containing the solution has not been found.

ERROR: The output results file cannot be opened.

ERROR: Writing error in the output results file.

Error messages. Spanish output:

ERROR: Selección fuera de límites.

ERROR: Valor no numérico.

ERROR: El módulo de elasticidad tiene que ser positivo.

ERROR: El esfuerzo último de tracción tiene que ser mayor que cero.

ERROR: El límite elástico no puede ser cero.

ERROR: El límite elástico tiene que ser menor que el esfuerzo último.

ERROR: The Poisson's modulus must be positive.

ERROR: The plastic deformation for ultimate stress must be greater than $(F_{tu}/F_{cy}) \cdot 0.2 \%$.

ERROR: El número de Ramberg-Osgood tiene que ser: $N_{ro} > 1$.

ERROR: El porcentaje de espesor del plaqueado tiene que ser: $0 < R_p < 100$.

ERROR: El espesor tiene que ser positivo.

ERROR: El paso tiene que ser positivo.

ERROR: El coeficiente de empotramiento tiene que ser positivo.

ERROR: No se encuentra intervalo que contiene la solución.

ERROR: No se puede abrir el fichero de salida de resultados.

ERROR: Error de escritura en el fichero de salida de resultados.

5 EXAMPLES

The case to be solved consists in a joint between panel and stringer, fastened each other, as shown in the following figure:

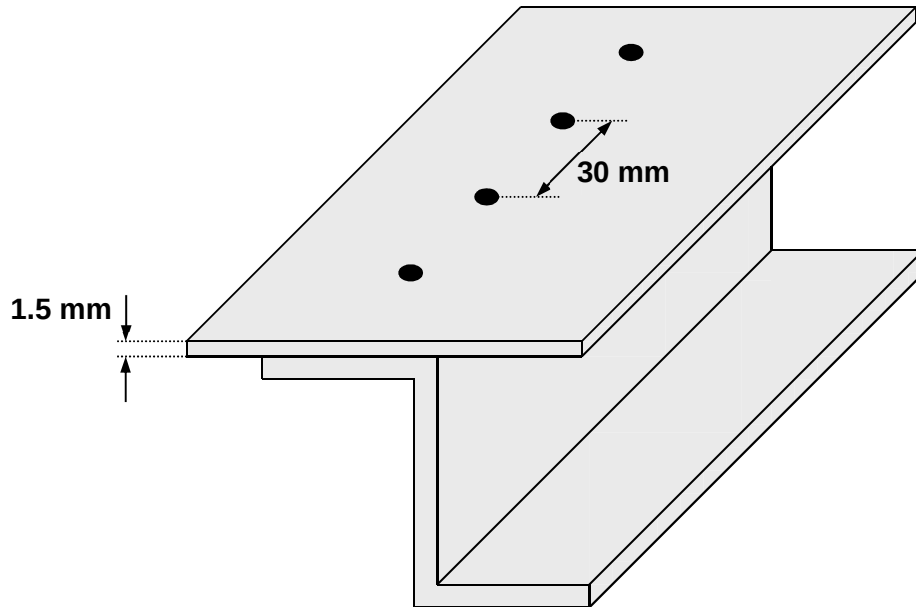


Figure 2. Example geometry.

The above panel will be analysed for inter-rivet buckling. The material will be assumed to be 7050 T7451, bare machined plate for example no. 1, and clad for example no. 2, with the following properties:

- Core material:

Young modulus in compression	E_c	=	73084.0 N/mm ²
Ultimate tensile stress	F_{tu}	=	524.0 N/mm ²
Yield stress in compression	F_{cy}	=	441.3 N/mm ²
Poisson's ratio	ν	=	0.33
Plastic deformation under ultimate tensile stress	e	=	10.0 %

- Clad material: (example 2 only)

Yield stress in compression of the cladding	F_{cy}	=	75.0 N/mm ²
Percentage of thickness of cladding	R_p	=	20.0 %

The fasteners have protruding head, that means a clamping coefficient of $C_c = 3.0$.

With all the above data, now it is possible to run the program for both examples. The screen data flow, with user inputs in bold type, as well as the output results files will be as follows:

Example 1. Bare machined plate. Data flow on screen:

```
*****
*                                     *
*          <<< F R E M >>>          *
*                                     *
*****
```

Idioma / Language:

=====

1 ... ESPAÑOL

2 ... ENGLISH

Idioma / Language: 2

```
*****
*                                     *
*   MATERIAL   *
*                                     *
*****
```

Material denomination 7050 T7451

TYPE OF MATERIAL:

=====

1 ... Homogeneous (unclad)

2 ... Clad

Type of material 1

```
*****
*                                     *
*   UNITS     *
*                                     *
*****
```

FORCE UNITS:

=====

1 ... N

2 ... daN

- 3 ... kN
- 4 ... kg
- 5 ... lb
- 6 ... klb

Force units 1

Example 1. Bare machined plate. Data flow on screen: (cont.)

LENGTH UNITS:

=====

- 1 ... mm
- 2 ... cm
- 3 ... m
- 4 ... in

Length units 1

```
*****
*                                     *
*   MECHANICAL PROPERTIES OF THE MATERIAL   *
*                                     *
*****
```

Young modulus in compression 73084

Ultimate tensile stress 524.0

Yield stress in compression 441.3

Poisson's modulus 0.33

Plastic deformation for ultimate stress (%) 10.0

Number of Ramberg-Osgood in compression 22.78

```
*****
*                                     *
*   FASTENING DATA   *
*                                     *
*****
```

Panel thickness at the fastened zone 1.5

Fasteners pitch 30.0

TYPE OF FASTENERS HEAD:

=====

- 1 ... Countersunk head
- 2 ... Protruding head
- 3 ... Traction head
- 4 ... Other kind of head

Type of fasteners head 2

Clamping coefficient of fasteners 3.0

Example 1. Bare machined plate. Data flow on screen: (cont.)

```
*****
*
*   ALLOWABLE STRESS FOR INTER-RIVET BUCKLING   *
*
*****
```

Buckling stress in plastic regime -374.0 N/mm²

Tangent modulus 60629.8 N/mm²

Do you want to print the results in an file? (Y/N): Y

Name of the output results file: Example1.txt

Example 1. Bare machined plate. Output results file:

```
*****
*
*   <<< F R E M >>>   *
*
*****
```

Input data:

=====

Material:

Denomination: 7050 T7451

Type of material: Homogeneous

Mechanical properties of the material:

Young modulus in compression: 73084.0 N/mm²
 Ultimate tensile stress: 524.0 N/mm²
 Yield stress in compression: 441.3 N/mm²
 Poisson's modulus: 0.33
 Plastic deformation for ultimate stress: 10.00 %
 Number of Ramberg-Osgood in compression: 22.78

Fastening data:

Panel thickness at the fastened zone: 1.50 mm
 Fasteners pitch: 30.00 mm
 Type of fasteners head: Protruding
 Clamping coefficient of fasteners: 3.000

Allowable stress for inter-rivet buckling:

=====

Buckling stress in plastic regime: -374.0 N/mm²
 Tangent modulus: 60629.8 N/mm²

Example 2. Clad plate. Data flow on screen:

```
*****
*                                     *
*      <<< F R E M >>>              *
*                                     *
*****
```

Idioma / Language:

=====

1 ... ESPAÑOL

2 ... ENGLISH

Idioma / Language: 2

```
*****
*                                     *
*      MATERIAL                      *
*                                     *
*****
```

Material denomination 7050 T7451

TYPE OF MATERIAL:

=====

- ```
1 ... Homogeneous (unclad)
2 ... Clad
```

Type of material ..... 2

\*\*\*\*\*

\* \*

\* UNITS \*

\* \*

\*\*\*\*\*

FORCE UNITS:

=====

- |   |     |     |
|---|-----|-----|
| 1 | ... | N   |
| 2 | ... | daN |
| 3 | ... | kN  |
| 4 | ... | kg  |
| 5 | ... | lb  |
| 6 | ... | k1b |

Force units ..... 1

**Example 2. Clad plate. Data flow on screen: (cont.)**

LENGTH UNITS:

=====

- 1 ... mm
- 2 ... cm
- 3 ... m
- 4 ... in

Length units ..... 1

\*\*\*\*\*

\* \*

\* MECHANICAL PROPERTIES OF THE CORE \*

\* \*

\*\*\*\*\*

Young modulus in compression ..... 73084

Ultimate tensile stress ..... 524.0

Yield stress in compression ..... 441.3  
 Poisson's modulus ..... 0.33  
 Plastic deformation for ultimate stress (%) ..... 10.0  
 Number of Ramberg-Osgood in compression ..... 22.78

```

* *
* MECHANICAL PROPERTIES OF THE CLADDING *
* *

```

Yield stress in compression ..... 75.0  
 Percentage of thickness of cladding ..... 20.0

```

* *
* FASTENING DATA *
* *

```

Panel thickness at the fastened zone ..... 1.5  
 Fasteners pitch ..... 30.0

## Example 2. Clad plate. Data flow on screen: (cont.)

TYPE OF FASTENERS HEAD:  
 =====

- 1 ... Countersunk head
- 2 ... Protruding head
- 3 ... Traction head
- 4 ... Other kind of head

Type of fasteners head ..... 2  
 Clamping coefficient of fasteners ..... 3.0

```

* *
* ALLOWABLE STRESS FOR INTER-RIVET BUCKLING *
* *

```

\*\*\*\*\*

Buckling stress in plastic regime ..... -310.8 N/mm<sup>2</sup>

Tangent modulus ..... 50387.1 N/mm<sup>2</sup>

Do you want to print the results in an file? .... (Y/N): Y

Name of the output results file: Example2.txt

## Example 2. Clad plate. Output results file:

\*\*\*\*\*  
\*  
\* <<< F R E M >>> \*  
\*  
\*\*\*\*\*

Input data:

=====

Material:

-----

Denomination ..... 7050 T7451

Type of material ..... Clad

Mechanical properties of the core:

-----

Young modulus in compression ..... 73084.0 N/mm<sup>2</sup>

Ultimate tensile stress ..... 524.0 N/mm<sup>2</sup>

Yield stress in compression ..... 441.3 N/mm<sup>2</sup>

Poisson's modulus ..... 0.33

Plastic deformation for ultimate stress .... 10.00 %

Number of Ramberg-Osgood in compression .... 22.78

Mechanical properties of the cladding:

-----

Yield stress in compression ..... 75.0 N/mm<sup>2</sup>

Percentage of thickness of cladding ..... 20.00 %

Fastening data:

-----

Panel thickness at the fastened zone ..... 1.50 mm

Fasteners pitch ..... 30.00 mm

Type of fasteners head ..... Protruding

---

Clamping coefficient of fasteners .....: 3.000

Allowable stress for inter-rivet buckling:

=====

Buckling stress in plastic regime .....: -310.8 N/mm<sup>2</sup>

Tangent modulus .....: 50387.1 N/mm<sup>2</sup>



## 6 APPENDICES

### 6.1 EFFECTIVE WIDTH OF SKIN UNDER STRAIGHT STRINGERS

The compression buckling stress of a supported plate/sheet of width  $b$  is:

$$\sigma_b = K_c \cdot E \cdot (t/b)^2 \quad (\text{Niu 11.2.2, Bruhn C7.13-14})$$

In a stiffened panel, the supports of the skin, the stringers, shall be able to sustain more compression stress beyond the skin local buckling. The sheet in the buckled state has the ability to approximately hold this stress as the stiffener continues to take load until it reaches the stiffener crippling stress. The effective skin width is defined to set a cross section area at the same stress as the stiffener equivalent to this stringer and the buckled skin sheet. The classical expression for this is:

$$w = t \cdot (K_c \cdot E / F_{st})^{0.5} \quad (\text{Niu 14.2.1})$$

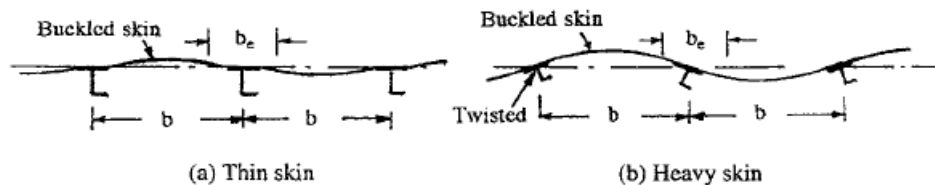


Figure 3. Thin and heavy skins (Niu)

When  $K_c$  is made equal to 3.62 (the “heavy skin” mode in Niu’s book), the familiar expression results:

$$w = t \cdot (3.6 \cdot E / F_{st})^{0.5} = 1.9 \cdot t \cdot (E / F_{st})^{0.5} \quad (\text{Bruhn C7.16})$$

Greater values of  $K_c$  are accepted by Niu from  $b/t$  ratios of 40 up to the “fixed edges” condition at  $b/t$  equal to 110 of “thin skins” (Niu’s Fig. 14.2.3). This potential improvement, which does not explicitly account for the actual stiffness of the stringer, shall not be applied. On the contrary... Newell suggested to change the coefficient 1.9 to 1.7 in configuration with light stringers. This later form has been widely used in the industry.

$$w = 1.7 \cdot t \cdot (E / F_{st})^{0.5}$$

For panels with different thickness at each side of the stringer, it will be practical to use the semi-effective width instead:

$$w_i = 0.85 \cdot t \cdot (E / F_{st})^{0.5}$$

It is remarked that the  $E$  in the formula is the Young’s modulus of the material, even when the stress of the stringer exceeds the yield stress (see discussion in Bruhn C7.12). Niu suggests to apply a plasticity correction factor to the Young modulus only for  $b/t$  ratios below 40. This is done in order to account for the reduction of the initial buckling stress due to plastic effects in narrow plates (with typical aluminum alloys used in aerospace, it is easy to check, with first formula in this page, that this approach is conservative).

Note that, as presented above, the effective width is function of the applied loads (the stress at the stringer/skin junction). It is reasonable to simplify this expression by conservatively use the value at

the stiffener failure stress, i.e. at the stiffener crippling stress, no matter what is the applied compression load. In this case the section inertia would be underestimated below the stringer crippling load, which will affect the calculation of stress distribution in the section, or the column buckling load.

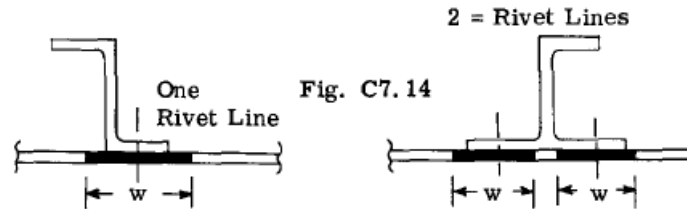


Figure 4. Effective width for sheet-stiffener units fastened by single attachment lines

Previous formulas are felt as appropriate for general design purposes. Literature presents also alternatives for other configurations, when the stiffener can be expected to provide support to the skin beyond the simple-support condition (see for example suggestions in Bruhn C7.12 or Niu 14.2, for integral stringers or staggered rivets rows).

### Effective width corrected with plasticity

As discussed in previous section, the effective width shall be calculated with the Young modulus corrected by plasticity only in narrow plates ( $b/t$  ratios below 40). The plasticity correction factor  $\eta$  applicable ( $\eta \cdot E$  replacing  $E$  in previous formulas) shall be that corresponding to buckling of flat plates in compression. Some classical expressions available for this problem are:

- Niu, section 11.2 (B)  $\eta = (E_t/E)^{0.5}$
- Bruhn, Fig. C5.8 & NACA TN 3781  $\eta = 0.5 \cdot (E_s/E) \cdot \{1 + 0.5 \cdot [1 + (3 \cdot E_t/E_s)]^{0.5}\} \cdot (1 - \nu_e^2)/(1 - \nu^2)$   
with  $\nu = \nu_p - (E_s/E) \cdot (\nu_p - \nu_e)$

Tangent and secant moduli in expressions above are function of the stress level in the material. In this case, in order to be consistent with the general approach and Niu's discussion,  $E_t$  and  $E_s$  values for obtaining  $\eta$  shall be computed at the local buckling stress level of the skin panel.

### Effective width corrected due to different stringer material

The sheet directly adjacent to the stiffener attachment line must undergo the same strain as the stiffener. In case the skin and the stringer are made of different materials, the correction for this condition can be made by multiplying previous value by  $F_{sh}/F_{st}$ :

$$w' = w \cdot (F_{sh} / F_{st}) \quad (\text{Bruhn, formula C7.18})$$

... where  $F_{sh}$  is the stress at the skin sheet for the strain level seen by the stringer.

Effective width corrected with inter-rivet buckling in skin

When the sheet between rivets buckles before the crippling stress of the stiffener is reached, the sheet in the buckled state has the ability to approximately hold this stress as the stiffener continues to take load until it reaches the stiffener crippling stress. This buckling sheet strength can be taken advantage of by reducing the effective sheet area. Thus effective width equals,

$$w'' = w' \cdot (F_{ir} / F_{st}) \quad (\text{Bruhn, formula C7.23})$$